



Retinal Review Workbench

Team Image Sampling Requirements

How the team plans and monitors retinal image coverage without adding clinician work

Software version	0.7.0 (no software change is implied by this document)
Document version	1.0 (2026-09-23)
Document ID	RRW-TEAM-SAMP-001
Audience	Project team, clinical lead, dataset coordinator, technical lead, hospital coordinator
Status	Draft for owner and clinical approval of the OPEN items (Chapter 9)
Source of truth	docs/clinician/sampling/REQUIREMENTS_TRACEABILITY.md
Companion	Clinician Image Selection Guide (RRW-CLIN-SEL-001)

Research and clinical-support proof of concept for public or synthetic data. Model output is assistive evidence; human review remains authoritative.

Table of contents

1	At a Glance	2
2	Data Ownership and Staging Boundary	3
3	Source Class: Refer and Not-refer	6
4	Secondary Coverage: Sex, Eyes, and Patients	8
5	Image Suitability	10
6	Optional Metadata Dimensions	12
7	Team Monitoring	12
8	What Clinicians Are Not Required to Do	14
9	Open Owner and Clinical Decisions	15
10	Image sampling requirements traceability	16

1 At a Glance

This document tells the project team how to plan and monitor the retinal images collected for the current proof-of-concept (POC) annotation batch. It is written for the team. Clinicians get a separate one-page *Clinician Image Selection Guide*.

Important – the one rule behind this document

The team needs dataset coverage. Clinicians must not become data managers. Coverage is planned and counted by the team. The clinician only chooses suitable images during normal work and stages them through the approved hospital process.

Table 1.1: Where to find each answer.

Question the team must answer	Where it is answered
Who chooses the images, and who never does?	Section 2 (IMG-SAMP-001 to 005)
How many Refer versus Not-refer?	Section 3 (IMG-SAMP-010 to 022)
Are both sexes represented?	Section 4.1 (IMG-SAMP-030 to 033)
Are both left and right eyes represented?	Section 4.2 (IMG-SAMP-040 to 043)
How much metadata is Unknown?	Section 4.3 and Section 7
Are we collecting duplicates or repeated cases?	Section 4.4 (IMG-SAMP-050 to 061)
Are the images usable retinal images?	Section 5 (IMG-SAMP-070 to 076)
Which dimensions are mandatory and which are best effort?	Table 1.3
When do we stop collecting for this POC?	Section 3.4 (IMG-SAMP-022)
Which items are software constraints, and which are planning only?	Chapter 10 (traceability register)

1.1 How to read a requirement

Every requirement has an IMG-SAMP- ID and one or more **levels**. The full wording, evidence, and approval status are in the traceability register (Chapter 10). That register is the single source of truth; this document explains it.

Table 1.2: Requirement levels.

Level	Meaning
SW	Implemented software behaviour on main, verified in code and tests.
OWN	Explicit owner/team decision.
TGT	Internal POC sampling target. Not a clinical requirement, not a prevalence claim, not a powered sample-size calculation.
EXT	External best-practice recommendation, adopted as guidance only.
OPEN	Proposed; needs owner or clinical approval before use (Section 9).

1.2 Dimensions at a glance

Table 1.3: Mandatory versus best-effort dimensions.

Dimension	Status for this POC	Who provides the value	Unknown allowed?
Source class Refer / Not - refer	Primary axis , required	Already held by the source workflow; recorded by the team	Not expected; see Section 3.2
Sex	Secondary coverage, best effort	Approved source metadata only	Yes
Laterality (left/right)	Secondary coverage, best effort	Confirm Image in the Workbench (existing step)	Yes
Patient grouping	Required for counting patients	Pseudonymous patient key in Confirm Image	Yes (unlinked)
Image suitability	Required for entry	Clinician glance (Use / Review first / Skip), then Workbench admission	Not applicable
Age, device, site, period	Optional, team only	Approved metadata when available	Yes; defer if unavailable

2 Data Ownership and Staging Boundary

2.1 Purpose

The Workbench turns images a clinician has already chosen into reviewed, traceable dataset records. It does not decide which hospital images are collected. This chapter fixes that boundary so that nobody on the team, and no future feature, drifts across it by accident.

2.2 The boundary

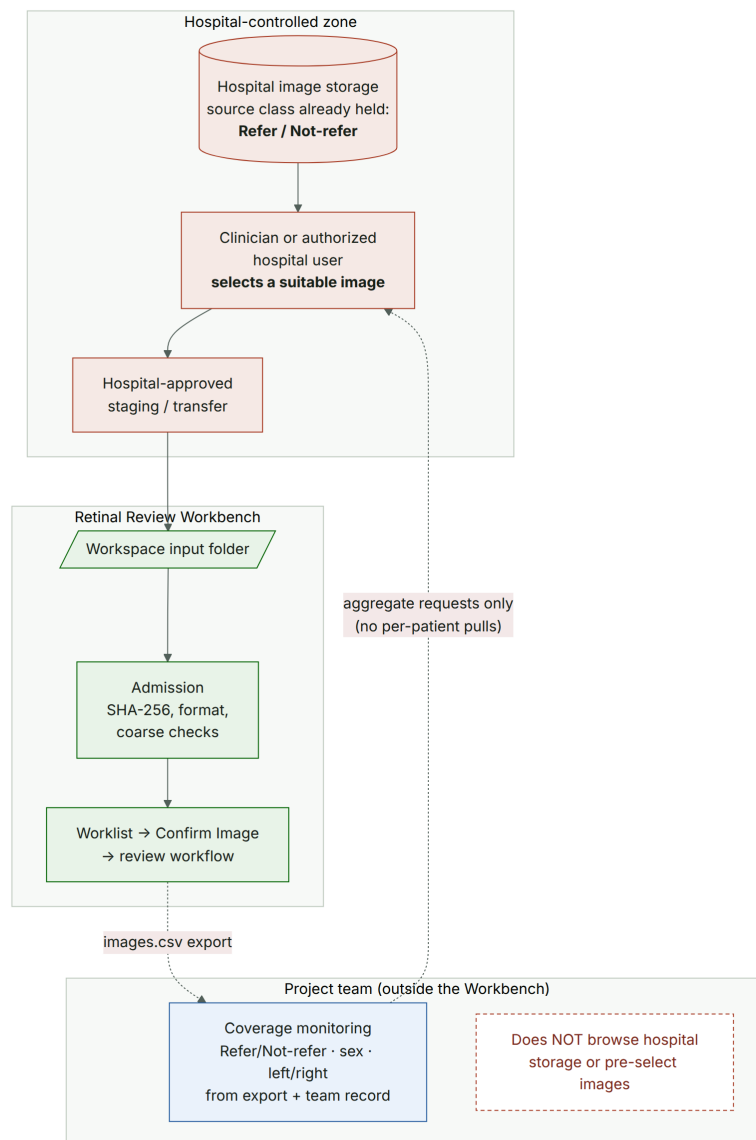


Figure 2.1: Image ownership and staging boundary. Source: docs/adr/architecture/03-image-staging-boundary.mmd.

1. **Hospital-controlled storage** holds the candidate retinal images. The source class Refer / Not-refer is already recorded there by the existing hospital workflow.
2. **A clinician or authorized hospital user** chooses a suitable candidate image during normal work (IMG-SAMP-002).
3. **The hospital-approved staging or transfer process** copies only that selected, original file into the Workspace input folder (IMG-SAMP-003, IMG-SAMP-004).
4. **The Workbench** admits what it finds in that folder: it records the SHA-256 source identity, checks the format, and runs coarse admission checks. The case then follows the normal Worklist → **Confirm Image** → review workflow.
5. **The project team** monitors coverage from the Workbench export and its own source-class record (Section 7). Gaps go back to the hospital coordinator only as aggregate requests (IMG-SAMP-092).

Safety – what the team never does

The project team and the Workbench do **not** browse, query, or harvest hospital-controlled storage, and do **not** pre-select, prioritise, or pull images for clinicians (IMG-SAMP-001). There is no automatic PACS query and no automatic patient selection. Do not propose one without a separate owner-approved specification.

2.3 What the software actually does at the boundary

These facts are verified in the current main branch. They describe the Workbench's behaviour; they do not add new obligations.

- The Workbench reads **only the configured Workspace input folder, top level only**. It does not follow sub-folders or symlinks (`scan_input_folder` in `dr_support/services/admission.py`).
- Ancillary files (`.csv`, `.json`, `.md`, `.txt`, `.log`, `.toml`, `.xml`, `.yaml`) in the input folder are ignored. They never become cases.
- Supported inputs are JPEG, PNG, TIFF, and single-frame ophthalmic-photography DICOM. Unsupported or unreadable files are reported, not silently dropped.
- Source files are never moved, renamed, rewritten, or recompressed. The SHA-256 identity of each source is recorded, and a changed file is flagged as `Source changed`.

2.4 Privacy at the staging boundary

- Stage the **original** file. Do not stage screenshots, crops, annotated copies, or re-exported derivatives (IMG-SAMP-004).
- Staged filenames and folder names must not carry direct identifiers. The export includes the `filename` column. Any pseudonymous naming is done by the approved hospital process, not invented image by image by the clinician (IMG-SAMP-005, **OPEN**).
- Team tracking records use pseudonymous keys and the Workbench `image_id` / `image_sha256` only. They are stored according to hospital policy and never committed to the repository (IMG-SAMP-091, IMG-SAMP-110).

3 Source Class: Refer and Not-refer

3.1 What the source class is

The source hospital classifies images only as **Refer** or **Not-refer**. This existing class is the **primary sampling axis** for the POC (IMG-SAMP-010). Clinicians do not create it at selection time; it already exists in the source workflow.

Important – two separate concepts

The source class (Refer / Not-refer) is **not** the Workbench clinician DR grade (0–4) and is not an annotation. The Workbench stores no source-class field. Its grade is recorded later, inside the review workflow, as AI_ACCEPTED, AI_CORRECTED, or MANUAL. Never map one onto the other, overwrite one with the other, or derive one from the other (IMG-SAMP-011).

At pre-selection the clinician is **not** asked for a DR grade, lesion labels, an image-quality score, a research cohort, a train/validation/test group, or demographic data (IMG-SAMP-012). Detailed review stays inside the Workbench.

3.2 Linking the source class to each staged image

Because the Workbench has no source-class field, the team must know which class each staged image came from without asking clinicians to type it per image. The mechanism is an **open owner decision** (IMG-SAMP-013). Candidate options, all of which work with the current software unchanged:

Table 3.1: Options for linking the source class to staged images (owner to choose).

Option	How it works	Clinician effort	Notes
A. One Workspace per class	Two Workspace profiles (for example <i>POC batch – Refer</i> and <i>POC batch – Not-refer</i>), each with its own input folder.	Stage into the folder for the class. No typing.	Uses existing Workspace profiles; each export is already split by class.
B. Hospital-supplied list	The approved staging process produces a pseudonymous list (file → class, and sex where approved) that goes to the team.	None	Needs hospital IT or coordinator support. Keep the list outside the input folder.
C. Coordinator batch log	Each staged batch is a single class, recorded by the coordinator.	None	The lightest option. It depends on batch discipline.

3.3 Primary sampling target

POC / annotation-coverage planning target (IMG-SAMP-020): collect Refer and Not-refer images in approximately equal numbers, with each class between **40 % and 60 %** of the batch.

Why approximately balanced:

- the DR-grade task needs both classes with enough examples to review;
- a small batch collected at natural prevalence could leave one class with very few examples;
- a band is more realistic than an exact 50/50, because images are chosen during normal work.

Table 3.2: POC / annotation-coverage planning example. This is **not** a statistically powered clinical-validation sample-size calculation.

Target total (N)	Refer (about)	Not-refer (about)	Acceptable band per class
50	25	25	20 to 30
100	50	50	40 to 60
200	100	100	80 to 120

Safety – balanced collection is not prevalence

A balanced batch does not represent the hospital population. Do not use it to estimate prevalence, sensitivity, specificity, or clinical performance (IMG-SAMP-021). If the purpose later becomes prevalence estimation or clinical-performance evaluation, the sampling design must be redone for that purpose. External reporting guidance (FDA draft guidance on AI-enabled devices, 2025; TRIPOD+AI, 2024) expects the data to be representative of the intended population *for that purpose*, with its characteristics reported.

Review of alternatives. Lesion annotations come mainly from Refer images. If the owner later decides that lesion-annotation volume matters more than grade-task balance, a Refer-leaning band (for example 55–65 % Refer) is defensible *as a planning choice*. Until then, the balanced 40–60 % band is the recommended default.

3.4 Stopping rule

Proposal (IMG-SAMP-022, TGT + OPEN). Stop collecting for the current batch when **all** of the following are true, or when the owner decides to stop:

1. the owner-approved total N has been reached;
2. each class is at least 40 % of N;
3. no more than 20 % of admitted images have Unknown laterality;
4. the monitoring summary (Section 7) has been reviewed and any flags have been noted to the owner.

Sex and laterality flags (Section 4) do **not** block stopping. They are reported, not enforced.

4 Secondary Coverage: Sex, Eyes, and Patients

Sex and laterality are **coverage dimensions**. They are not per-image clinician tasks and not quotas. The team counts them. Clinicians are never asked to balance them.

4.1 Sex

- Record sex **only** when it is already available from approved source metadata or the approved workflow. Clinicians never re-enter sex per image (IMG-SAMP-030). The Workbench stores no sex field.
- Both sexes should be represented where data are available. Report Male / Female / Unknown counts by source class (IMG-SAMP-031).
- **Soft monitoring flag (internal POC target):** note it to the owner if either known sex is below 30 % of known-sex images (IMG-SAMP-032). This is not a clinical requirement, not a prevalence claim, and not a reason to exclude an image.

The 30 % threshold is operational. It exists to make marked imbalance visible early. A strict 50/50 is not assumed to be scientifically correct: the sex distribution of people with diabetes attending the source service may itself be unequal.

4.2 Left and right eyes

- Both left and right eyes should be represented. Report the laterality distribution by source class (IMG-SAMP-040). There is no fixed 50/50 quota.
- **Count only the clinician-confirmed eye** from **Confirm Image**. In the export this is laterality with `laterality_resolution_method = MANUAL` (IMG-SAMP-041).
- Filename patterns, optional OCR, and DICOM *Image Laterality* (0020,0062) are **candidates only**. The Workbench extracts the DICOM value as a metadata candidate, but the resolver does not use it. The clinician's **Confirm Image** choice is authoritative inside the workflow.
- Never guess laterality from a filename for team statistics. Unconfirmed or unsafe laterality stays Unknown (IMG-SAMP-042).
- **Soft monitoring flag:** note it if either known eye is below 30 % of known-laterality images (IMG-SAMP-043).

4.3 Unknown values

Unknown / *not available* is an acceptable value for sex, laterality, and every optional dimension (IMG-SAMP-033). Do not exclude a useful retinal image only because demographic metadata is missing, unless the owner or clinical team explicitly requires it. Report the Unknown rate in every monitoring update, so that missing data is visible rather than hidden (STANDING Together, 2025).

4.4 Same patient, both eyes

- Several images from the same patient, including both eyes, are allowed. They are **not independent patients**. Preserve the pseudonymous patient key recorded in **Confirm Image**, and report **distinct patients** as well as images (IMG-SAMP-050).
- The export already carries `patient_key` and `training_group_key` (`patient:<key>`), and states `grouping_policy: patient_key_only`; no automatic train-validation-test split.
- **Future safeguard.** Any later ML train/validation/test experiment should split at patient or group level where identity is available (IMG-SAMP-051). Splitting by image can inflate reported accuracy substantially when one patient's images land in both sets (Tampu et al., 2022). This POC does **not** add a split; the safeguard is recorded here for future experiment designs only.
- **Soft monitoring flag:** note any patient contributing more than 4 images to a batch (IMG-SAMP-052, value to be approved).

4.5 Duplicates and repeated images

Table 4.1: Duplicate handling.

Situation	Detected by the Workbench?	What happens
The same file staged twice, or byte-identical copies	Yes , by SHA-256	One case with several aliases, shown as <code>Duplicate content</code> (IMG-SAMP-060).
The same file changed after admission	Yes	Source changed integrity status.
A repeat capture of the same eye, or a resized or re-exported copy	No	The team reviews repeated patient + eye combinations (IMG-SAMP-061).

Clinicians only need to avoid *knowingly* staging the same image twice. They are not asked to search for duplicates.

5 Image Suitability

5.1 A glance-level decision, not a score

Clinicians make a three-way decision for each candidate image: **Use**, **Review first**, or **Skip** (IMG-SAMP-070). There is no numeric quality score, no checklist, and no form. The wording follows established concepts: the NHS Diabetic Eye Screening Programme's *adequate* / *inadequate* image definitions, and the *Good* / *Usable* / *Reject* categories of the EyeQ quality dataset. It is kept deliberately simpler than both.

Table 5.1: The clinician suitability decision. Clinical wording requires sign-off.

Decision	Meaning	Examples
Use	A colour fundus/retinal photograph that already has a Refer or Not-refer class, is the original file, and shows enough retina for normal reading: optic disc, macula, and vessels visible.	A typical screening photograph (IMG-SAMP-071).
Review first	Might be useful, but something is uncertain. Take a second look. If you would still use it clinically, stage it without inventing metadata. Otherwise skip it or ask the coordinator.	Borderline blur or exposure; uncertain field or eye; possible duplicate; not sure enough retina is visible (IMG-SAMP-072).
Skip	Not suitable for this collection.	Wrong modality (OCT, external-eye or anterior-segment photograph); corrupt or unreadable file; screenshot, crop, or annotated copy; non-retinal image; multi-frame or non-ophthalmic DICOM (IMG-SAMP-073).

For reference, the NHS programme defines an adequate macula-centred image as one in which “the centre of fovea is > 2DD from edge of image and the 3rd generation vessels are visible”. An adequate disc-centred image shows the complete optic disc more than 2 disc diameters from the edge, with fine vessels visible on the disc surface. These definitions inform the **Use** wording; clinicians are **not** asked to measure them.

5.2 Suitability is not the source class

Safety – do not relabel poor images

Image suitability and gradability are separate from Refer / Not-refer. A poor-quality or ungradable image must never be relabelled Refer or Not-refer to fit a sampling target (IMG-SAMP-074).

This separation matches established practice:

- In NHS screening, an inadequate image set with no referable features gets its own **ungradable (U)** outcome and an inadequate-image pathway. Referable features seen on an inadequate image are still referred at their disease grade.
- The FDA De Novo summary for IDx-DR (DEN180001) describes a separate “**insufficient quality**” output, in addition to its two disease outputs.

5.3 What the Workbench already does

- **Admission** runs coarse screens: minimum dimension, extreme aspect ratio, near-uniform image, extreme brightness, and weak colour/field signals. An image that fails a screen is flagged **Needs review** or **Image quality review**, not rejected. These checks are safety screens, **not** a gradability assessment (IMG-SAMP-076).
- **Admission quality state** (GRADABLE, UNGRADABLE, NEEDS_REVIEW, NOT_EVALUATED) is independent of the grade. UNGRADABLE blocks DR-ready and Lesion-ready in the dataset export.
- A staged image later found ungradable keeps the existing Workbench quality state (**Image quality issue**, recorded through the existing QUALITY_INADEQUATE admission action). **No new clinical label is created** (IMG-SAMP-075). The team counts these images. Whether they count toward N is an owner decision.

6 Optional Metadata Dimensions

Regulatory and reporting guidance asks developers to characterise their datasets. The FDA draft guidance on AI-enabled device software (January 2025) lists demographics, data acquisition equipment and conditions, and sites. STANDING Together (2025) asks for transparent documentation of who is represented and what is missing. These are **recommendations for how a dataset is described**, not quotas for this POC (IMG-SAMP-080).

Table 6.1: Optional dimensions. None becomes per-image clinician entry.

Dimension	Why it can matter	POC decision	How it would be collected
Age band	Subgroup characterisation	Optional; defer if not available	Approved source metadata only
Camera / device	Model robustness across devices	Optional; worth tracking when feasible	Metadata or batch-level note by the team, never per image by the clinician
Site / acquisition setting	Generalisability	Optional; one site expected	Batch-level note by the team
Capture period	Drift and era effects	Optional	Batch-level note by the team
Image quality / gradability	Model behaviour degrades with quality (EyeQ)	Covered by suitability and admission state (Section 5)	Existing Workbench admission state

Rule. If a dimension cannot be obtained without extra clinician data entry, defer it and state “not collected” in the dataset description. Do not add it to the clinician guide.

7 Team Monitoring

7.1 Two simple summaries

The team keeps **two small summaries**, not an 18-cell quota matrix (IMG-SAMP-090). If joint counts are ever needed (for example sex × eye × class), generate them programmatically from the same data.

Table 7.1: Summary 1 – sex by source class.

Sex	Refer	Not-refer	Total
Male	<i>n</i>	<i>n</i>	<i>n</i>
Female	<i>n</i>	<i>n</i>	<i>n</i>
Unknown	<i>n</i>	<i>n</i>	<i>n</i>
Total	<i>n</i>	<i>n</i>	<i>N</i>

Table 7.2: Summary 2 – confirmed laterality by source class.

Eye (confirmed)	Refer	Not-refer	Total
Left	<i>n</i>	<i>n</i>	<i>n</i>
Right	<i>n</i>	<i>n</i>	<i>n</i>
Unknown	<i>n</i>	<i>n</i>	<i>n</i>
Total	<i>n</i>	<i>n</i>	<i>N</i>

Each update also reports these metrics:

Table 7.3: Monitoring metrics per update.

Metric	Source
Refer share of N, and whether it is inside the 40–60 % band	Team source-class record
Unknown rate for sex and for laterality	Team record; images.csv laterality, laterality_resolution_method
Distinct patients; images per patient; patients above the flag value	images.csv patient_key
Duplicate content (aliases)	Worklist status / admission integrity
Suitability outcomes: Needs review, Image quality issue, Cannot analyze, Excluded	images.csv modality_admission, quality_state, queue_state
DR-ready and Lesion-ready counts	manifest.json counts
Flags raised (sex, eye, per-patient cap)	The thresholds in Section 4

7.2 How to produce the summaries

1. In the Workbench **Datasets** page, choose **Export manifest**. The export folder contains manifest.json, images.csv, and annotations.csv (schema s4.dataset-manifest.v2, see docs/reference/DATASET_MANIFEST.md).
2. Join images.csv to the team source-class record by image_id or image_sha256, using the mechanism chosen in Section 3.2 (IMG-SAMP-091).
3. Count laterality only when laterality_resolution_method is MANUAL. Otherwise count it as Unknown.
4. Store the output according to hospital policy. It contains no direct identifiers, and it is **not** committed to this repository.
5. Send any coverage gap to the hospital coordinator as an **aggregate** request only, for example “about 15 more Not-refer images”, never as a list of patients (IMG-SAMP-092).

A minimal reference implementation that uses only the Python standard library:

```
import csv
from collections import Counter

def summaries(images_csv, team_csv):
    """team_csv columns: image_sha256, source_class, sex."""
    with open(team_csv, encoding="utf-8") as f:
        team = {r["image_sha256"]: r for r in csv.DictReader(f)}
    sex, eye, patients = Counter(), Counter(), Counter()
    with open(images_csv, encoding="utf-8") as f:
        for row in csv.DictReader(f):
            rec = team.get(row["image_sha256"], {})
```

```

cls = rec.get("source_class") or "Unlinked"
sex[(rec.get("sex") or "Unknown", cls)] += 1
manual = row["laterality_resolution_method"] == "MANUAL"
known = row["laterality"] in ("LEFT", "RIGHT")
side = row["laterality"].title() if manual and known else "Unknown"
eye[(side, cls)] += 1
if row["patient_key"]:
    patients[row["patient_key"]] += 1
return sex, eye, patients

```

Unlinked in the output means that an exported image has no team source-class entry. Resolve that before reporting.

7.3 Worked example (illustrative numbers only)

Table 7.4: Illustrative laterality summary for N = 100.

Summary 2	Refer	Not-refer	Total
Left	21	19	40
Right	24	22	46
Unknown	7	7	14
Total	52	48	100

Reading of this example: Refer is 52 % (inside the band). Unknown laterality is 14 % (at most 20 %, so the stopping rule is met). Left is 47 % of known eyes (no flag). The batch can stop once the owner confirms N.

8 What Clinicians Are Not Required to Do

This list is an acceptance criterion (IMG-SAMP-100). Reject any process change that brings one of these tasks back to clinicians.

Table 8.1: Clinician burden boundaries.

Clinicians are not required to...	Who handles it instead
calculate quotas or balance Refer / Not-refer	Team monitoring (Section 7)
count male/female patients or left/right eyes	Team monitoring
enter demographic data already available elsewhere	Approved source metadata, or Unknown
assign a DR grade (0–4) before Workbench review	The Workbench Confirm DR Grade step, later
assign lesion labels before Workbench review	The Workbench Edit Annotations step, only when needed
score image quality numerically	A glance-level Use / Review first / Skip decision
choose train/validation/test groups	Not done in this POC (IMG-SAMP-051)
complete a per-image selection form	No form exists (IMG-SAMP-101)
search for duplicates	Workbench SHA-256 detection plus team review

What the clinician does: choose a suitable image in normal work, and stage it through the approved process. If unsure, do not invent metadata: choose Unknown, skip the image, or ask the coordinator. Everything after that happens in the existing Workbench workflow, which the clinician already uses (*Clinician User Manual*).

Expected extra effort: a few seconds per image beyond normal image choice. The only exception is a one-time read of the one-page guide.

9 Open Owner and Clinical Decisions

These items are proposals. They stay **OPEN** until the owner (and, where marked, the clinical lead) approves them. The defaults below are suggestions, not decisions.

Table 9.1: Open decisions.

#	Decision	Proposed default	Requirement	Approver
1	Total N for the current POC batch	100	IMG-SAMP-020, 022	Owner
2	Refer / Not-refer planning band	40–60 % per class	IMG-SAMP-020	Owner
3	How the source class (and sex) is linked to each staged image	Option A, one Workspace per class	IMG-SAMP-013	Owner, hospital coordinator
4	Hospital-approved staging process and pseudonymous filename convention	Defined by hospital IT	IMG-SAMP-003, 005	Owner, hospital IT
5	Soft flag thresholds for sex and laterality	30 % of known values	IMG-SAMP-032, 043	Owner
6	Images-per-patient flag	More than 4 per batch	IMG-SAMP-052	Owner
7	Whether ungradable or borderline staged images count toward N	Count separately; not toward N	IMG-SAMP-075	Owner, clinical lead
8	Clinical wording of Use / Review first / Skip	As in the Clinician Image Selection Guide	IMG-SAMP-070 to 073	Clinical lead
9	Optional dimensions to record (device, site, age band, period)	Device and site at batch level only	IMG-SAMP-080	Owner
10	Storage location and retention of team monitoring records	Per hospital policy	IMG-SAMP-091	Owner, hospital IT
11	Named dataset coordinator	To be named	IMG-SAMP-090, 092	Owner

None of these decisions requires a software change. If a decision would require one (for example a source-class field in the Workbench), it needs a separate normative specification under docs/ and owner approval before implementation, in line with AGENTS.md.

10 Image sampling requirements traceability

This file is the source of truth for every IMG-SAMP-* requirement. The Team Image Sampling Requirements PDF, the Clinician Image Selection Guide, the product presentation, and the technical briefing cite these IDs and do not restate them with different wording.

Changing a requirement here does **not** change Workbench admission, review, or export behaviour. Software behaviour changes need a separate owner-approved spec (see AGENTS.md).

10.1 How to read the columns

- **Level**
 - SW: implemented software behaviour on main, verified in code and tests.
 - OWN: explicit owner/team decision (V4 handoff).
 - TGT: internal POC sampling target. This is not a clinical requirement, not a prevalence claim, and not a powered sample-size calculation.
 - EXT: external best-practice recommendation. It is adopted only as guidance.
 - OPEN: proposed; needs owner or clinical approval before it is used.
- **Audience**
 - C: clinician or authorized hospital user.
 - T: project team or dataset coordinator.
 - D: developer.
 - H: hospital coordinator or IT.
- **Software-enforced?** says whether the Workbench itself enforces the requirement:
 - Yes: it enforces it.
 - Partial: it gives evidence but does not enforce it.
 - No: it plays no part.
- **Clinician action?** and **Team action?** name what each role actually does. None means nothing.

10.2 Requirements

ID	Requirement	Audience Level		Source type	Software-enforced? Clinician action?		Team action?	Owner approval needed?
IMG-SAMP-001	The project team and the Workbench do not browse, query, or harvest hospital-controlled storage, and do not pre-select images for clinicians.	T, H, D	OWN	Owner decision	Partial: the Workbench only scans its configured Workspace input folder	None	Do not request storage access; do not propose automatic PACS/storage queries	No (decided)
IMG-SAMP-002	A clinician or authorized hospital user chooses each candidate retinal image from hospital-controlled storage during normal work.	C	OWN	Owner decision	No	Choose the image	None	No (decided)
IMG-SAMP-003	Only selected images reach the Workbench, and only through the hospital-approved staging/transfer process into the Workspace input folder.	C, H	OWN	Owner decision	Partial: admission reads only the Workspace input folder, top level, no sub-folders	Stage through the approved process	Confirm the approved process exists before collection starts	Yes: the approved staging process itself
IMG-SAMP-004	Stage the original image file. Do not crop, annotate, recompress, convert, or screenshot it before staging.	C, H	OWN + SW	Owner decision; repository immutability rule	Partial: SHA-256 source identity, format/extension checks, source-change detection	Copy the original file	Investigate Source changed or format-mismatch cases	No
IMG-SAMP-005	Staged filenames and folders must not carry direct identifiers. Any pseudonymous naming is done by the approved hospital process, not invented per image by the clinician.	H, T	OPEN	Privacy boundary; export includes filename	No	None beyond the approved process	Agree a naming convention with the hospital	Yes
IMG-SAMP-010	The only source classification is Refer / Not-refer, as already held by the source hospital workflow. It is the primary sampling axis.	T, C	OWN	Owner decision	No: the Workbench stores no source class	None: the class already exists in the source workflow	Record the class for each staged image in the team tracking record	No (decided)
IMG-SAMP-011	Source class (Refer/Not-refer) is a separate concept from the Workbench clinician DR grade (0–4) and annotations. Neither is mapped onto, overwritten by, or derived from the other.	T, D	OWN + SW	Owner decision; code	Yes: grade values are 0–4 and grade sources are AI_ACCEPTED / AI_CORRECTED / MANUAL; no source-class field exists	None	Keep source class in the team record only	No
IMG-SAMP-012	During pre-selection the clinician is not asked for a DR grade, lesion labels, an image-quality score, a research cohort, a train/validation/test group, or demographic data.	C	OWN	Owner decision (minimum burden)	No	None	Keep forms and requests free of these fields	No (decided)

ID	Requirement	Audience Level		Source type	Software-enforced? Clinician action?		Team action?	Owner approval needed?
IMG-SAMP-013	How the source class (and sex, where used) is linked to each staged image is decided before collection, without adding per-image clinician data entry. Options: one Workspace per class, a hospital-supplied pseudonymous list, or coordinator batch logs.	T, H	OPEN	Gap: the Workbench has no source-class field	No	At most, stage into the folder for that class (only if that option is chosen)	Propose and run the chosen mechanism	Yes
IMG-SAMP-020	For the current POC annotation batch, aim for approximately balanced Refer / Not -refer collection (planning band 40–60 % per class).	T	TGT	POC / annotation-coverage planning target	No	None	Monitor counts; ask the coordinator for more of the under-represented class	Yes: target N and band
IMG-SAMP-021	A balanced collection is not hospital prevalence. Do not report prevalence, sensitivity/specificity, or clinical performance from this set without a new, purpose-specific sampling design.	T	EXT + OWN	FDA AI-enabled device draft guidance; TRIPOD+AI	No	None	Label every report “POC annotation-coverage sample”	No
IMG-SAMP-022	Stop collecting for the current batch when the owner-approved total N is reached, each class has at least its floor (default proposal: 40 % of N), and the Unknown-laterality share is at most 20 %, or when the owner decides to stop.	T	TGT + OPEN	Internal planning	No	None	Check the stopping rule at each monitoring update	Yes
IMG-SAMP-030	Sex is recorded only when it is already available from approved source metadata or the approved workflow. Clinicians never re-enter it per image.	T, C	OWN	Owner decision	No: the Workbench stores no sex field	None	Take sex from approved metadata only, or leave it Unknown	No (decided)
IMG-SAMP-031	Both sexes should be represented where data are available. Counts of Male / Female / Unknown are reported by source class.	T	OWN	Team requirement	No	None	Produce the sex × source-class table	No
IMG-SAMP-032	Soft monitoring flag: raise a note to the owner if either known sex is below 30 % of known-sex images. This is a flag, not a quota and not a reason to exclude an image.	T	TGT	Internal POC target	No	None	Flag it in the monitoring update	Yes: threshold
IMG-SAMP-033	Unknown / not available is an acceptable value for sex, laterality, and optional metadata. A useful retinal image is not excluded only because demographic metadata is missing.	T, C	OWN	Owner decision	Partial: laterality UNKNOWN is a valid Confirm Image value	Choose Unknown instead of guessing	Report Unknown rates	No
IMG-SAMP-040	Both left and right eyes should be represented. The laterality distribution is reported by source class. There is no fixed 50/50 quota.	T	OWN	Team requirement	No	None	Produce the laterality × source-class table	No
IMG-SAMP-041	Team laterality counts use the clinician-confirmed value from Confirm Image (<code>laterality_resolution_method = MANUAL</code> in <code>images.csv</code>). Filename, OCR, or DICOM candidates are evidence only.	T, D	SW + OWN	Code: Confirm Image contract and export columns	Yes: Confirm Image stores LEFT / RIGHT / UNKNOWN with method MANUAL	Set the eye in Confirm Image (the existing workflow step)	Count only confirmed values; treat others as Unknown	No

ID	Requirement	Audience Level		Source type	Software-enforced? Clinician action?		Team action?	Owner approval needed?
IMG-SAMP-042	Laterality is never guessed from a filename for team statistics. Unconfirmed or unsafe laterality stays Unknown.	T, C	OWN + EXT	Owner decision; DICOM Image Laterality is the standard attribute	Partial: resolver candidates need confirmation in the workflow	Choose Unknown when not sure	Do not back-fill from filenames	No
IMG-SAMP-043	Soft monitoring flag: note it if either known eye is below 30 % of known-laterality images.	T	TGT	Internal POC target	No	None	Flag it in the monitoring update	Yes: threshold
IMG-SAMP-050	Several images from one patient, including both eyes, are allowed. They are not independent patients: the pseudonymous patient key is preserved and the team reports distinct patients as well as images.	T, D	SW + EXT	Code: patient_key, training_group_id, Tampu 2022	Partial: grouping key exported; no automatic split	Link the patient key in Confirm Image (existing step)	Report distinct-patient count and images per patient	No
IMG-SAMP-051	Any future ML train/validation/test experiment splits at patient/group level where identity is available. This POC creates no split.	T, D	EXT + SW	GMLP principles; Tampu 2022; export grouping_policy	Yes: export states “no automatic train-validation-test split”	None	Apply this only in future experiment designs	No
IMG-SAMP-052	Soft monitoring flag: note any patient who contributes more than 4 images to a batch.	T	TGT + OPEN	Internal POC target	No	None	Flag it in the monitoring update	Yes: cap value
IMG-SAMP-060	Byte-identical files are detected by SHA-256 and shown as one case with aliases (Duplicate content).	T, D	SW	Code: admission integrity	Yes	None	Review duplicate aliases before counting	No
IMG-SAMP-061	Near-duplicates (repeat captures, re-exported or resized copies) are not detected automatically. The team reviews repeated patient + eye combinations. Clinicians avoid knowingly staging the same image twice.	T, C	SW + OWN	Code gap; team policy	No	Skip an image you know was already staged	Review repeated patient/eye combinations	No
IMG-SAMP-070	The clinician uses a three-way decision for each candidate: Use , Review first , or Skip . There is no score and no form.	C	OWN + EXT	Owner decision; NHS DESP adequacy concept; EyeQ Good/Usable/Reject	No	A glance-level decision	Maintain the guide	Yes: clinical sign-off of wording
IMG-SAMP-071	Use : a colour fundus/retinal photograph that already has a Refer/Not-refer class, in which the retina (optic disc, macula, vessels) is visible enough for normal reading, and which is the original file in a supported format.	C	EXT + SW	NHS DESP image-adequacy definitions; supported formats in code	Partial: format support and admission checks	Pick images like this	None	Yes: clinical sign-off
IMG-SAMP-072	Review first : borderline blur or exposure, uncertain field or laterality, a possible duplicate, or not sure enough retina is visible. Take a second look. If you would still use it clinically, stage it without inventing metadata. Otherwise skip it or ask the coordinator.	C	OWN	Owner decision	Partial: admission may flag Needs review / Image quality review	Second look only	Answer questions; track borderline outcomes	Yes: clinical sign-off

ID	Requirement	Audience Level		Source type	Software-enforced? Clinician action?		Team action?	Owner approval needed?
IMG-SAMP-073	Skip: wrong modality (for example OCT or an external-eye photo), corrupt or unreadable files, screenshots or edited/annotated derivatives, non-retinal images, and objects the Workbench cannot admit (for example multi-frame or non-ophthalmic DICOM).	C	SW + OWN	Code: unsupported modality/format handling	Partial: unsupported inputs are rejected or flagged	Do not stage	None	No
IMG-SAMP-074	Image suitability or gradability is a separate concept from source class. A poor-quality or ungradable image is never relabelled Refer or Not-refer to fit a target.	C, T	OWN + EXT	Owner decision; NHS DESP ungradable outcome; FDA DEN180001 "insufficient quality" output	Yes: admission quality_state is independent of grade; UNGRADABLE blocks DR-ready	None	Report suitability outcomes separately	No
IMG-SAMP-075	Staged images later found ungradable keep the existing Workbench quality state (Image quality issue / UNGRADABLE). No new clinical label is created. The team reports how many there are.	T, D	SW + OWN	Code: admission review actions	Yes	Existing admission action only if needed	Count them; decide with the owner whether they count toward N	Yes: counting rule
IMG-SAMP-076	Workbench automatic admission checks (size, aspect ratio, near-uniformity, extreme brightness, colour/field signals) are coarse safety screens, not a gradability assessment.	T, D	SW	Code: _classify_decoder only)	Yes (as a screen only)	None	Do not report them as quality grading	No
IMG-SAMP-080	Optional dimensions (age band, camera/device, site, capture period) are team-only. Collect them from approved metadata when this is feasible; otherwise defer them. They never become per-image clinician entry.	T	EXT + OWN	FDA AI-enabled device draft guidance; STANDING Together	No	None	Record them when available, or state them as not collected	Yes: which dimensions
IMG-SAMP-090	The team keeps two simple monitoring summaries: sex × source class and laterality × source class. Each update also reports Unknown rates, distinct patients, duplicates, and suitability outcomes.	T	OWN	Team requirement	No	None	Update after each staged batch	No
IMG-SAMP-091	Monitoring tables are produced by the team from the Workbench export (images.csv) joined to the team source-class record by image_id / image_sha256. They contain no direct identifiers and are not committed to this repository.	T, D	OWN + OPEN	Team process; export schema	No	None	Run the join; store the output per hospital policy	Yes: storage location
IMG-SAMP-092	Coverage gaps go back to the hospital coordinator only as aggregate requests (for example "more Not-refer images this week"), never as per-patient pull lists.	T, H	OWN	Owner decision	No	Occasional aggregate request, at batch level	Send aggregate requests only	No
IMG-SAMP-100	Clinicians are not required to calculate quotas, balance subgroups, count sex or eyes, enter demographics, assign a DR grade or lesions before review, score quality, choose research splits, or complete per-image forms.	C	OWN	Owner decision (acceptance criterion)	No	None	Reject any process change that adds these	No (decided)

ID	Requirement	Audience Level		Source type	Software-enforced? Clinician action?		Team action?	Owner approval needed?
IMG-SAMP-101	The clinician-facing guide stays at one page (two at most), has no form, calculator, or data entry, and states that the team monitors coverage.	C, T	OWN	Owner decision	No	Read it once	Keep it minimal; send proposed additions to the owner	Yes: for any added field
IMG-SAMP-110	Guides, presentations, monitoring tables, screenshots, and this repository never contain PHI, raw DICOM identifiers, or hospital storage paths.	All	OWN + SW	Security and privacy rules; export uses pseudonymous keys	Partial: pseudonymous patient_key validation; export has no absolute paths	None	Check before sharing	No

10.3 Evidence

ID	Repository evidence	External source
IMG-SAMP-001	dr_support/services/admission.py scan_input_folder (configured folder only); dr_support/services/workspaces.py	Owner decision (V4)
IMG-SAMP-002	None: happens outside the software	Owner decision (V4)
IMG-SAMP-003	scan_input_folder: top-level files only, symlinks and ancillary .csv/.json/.md/.txt skipped	Owner decision (V4)
IMG-SAMP-004	dr_support/imaging/contracts.py IntegrityStatus; tests/test_integrity.py; AGENTS.md immutability rule	None
IMG-SAMP-005	docs/reference/DATASET_MANIFEST.md (filename column); docs/operations/SECURITY_PRIVACY.md	None
IMG-SAMP-010	No Refer/Not-refer field in dr_support/** or frontend/src/**	Owner decision (V4)
IMG-SAMP-011	dr_support/workflow.py Review (grade 0–4), grade_review_source; docs/reference/DATASET_MANIFEST.md	NHS DESP grading definitions [E2] (grading and referral are separate schemes)
IMG-SAMP-012	None	Owner decision (V4)
IMG-SAMP-013	dr_support/services/workspaces.py (several Workspace profiles are supported)	Datasheets for Datasets [E10] (record the collection process)
IMG-SAMP-020	None	FDA AI-enabled device draft guidance [E7] (characterise the data); owner decision
IMG-SAMP-021	None	[E7]; TRIPOD+AI [E9]
IMG-SAMP-022	images.csv laterality column	Internal planning
IMG-SAMP-030	No sex field in contracts	[E7] and STANDING Together [E8] (report demographics when available)
IMG-SAMP-031	None	[E7]; [E8]
IMG-SAMP-032	None	Internal
IMG-SAMP-033	ConfirmImage.laterality accepts UNKNOWN (dr_support/workflow.py)	[E8] (state missing data openly)
IMG-SAMP-040	images.csv laterality	DICOM PS3.3 C.8.17.5 Image Laterality [E5]

ID	Repository evidence	External source
IMG-SAMP-041	dr_support/workflow.py confirm_image; dr_support/services/dataset.py IMAGE_FIELDS	[E5]
IMG-SAMP-042	dr_support/services/resolver.py (filename/OCR candidates); dr_support/imaging/dicom.py _laterality_candidate (extracted as a metadata candidate; not used by the resolver)	[E5]
IMG-SAMP-043	None	Internal
IMG-SAMP-050	dr_support/services/dataset.py training_group_key, grouping_policy	Tampu et al. 2022 [E11]
IMG-SAMP-051	manifest.json grouping_policy	GMLP guiding principles [E6]; [E11]
IMG-SAMP-052	None	Internal
IMG-SAMP-060	dr_support/services/admission.py _apply_integrity (DUPLICATE_CONTENT)	None
IMG-SAMP-061	None: no perceptual-hash detection is implemented	None
IMG-SAMP-070	None	NHS DESP image-quality guidance [E1]; EyeQ [E4]
IMG-SAMP-071	SUPPORTED_INPUT_TYPES in dr_support/services/admission.py	[E1] (macula- and disc-centred field adequacy)
IMG-SAMP-072	clinician_view labels in dr_support/services/admission.py	[E1]; [E4] (“Usable” category)
IMG-SAMP-073	DICOM_UNSUPPORTED_MODALITY, REJECTED_INVALID handling	DICOM Ophthalmic Photography IOD [E5]
IMG-SAMP-074	AdmissionMetadata.quality_state; dataset_case_eligibility (UNGRADABLE)	[E1] (ungradable outcome); FDA DEN180001 [E3]
IMG-SAMP-075	AdmissionReviewAction QUALITY_INADEQUATE (dr_support/contracts/admission.py)	[E1]
IMG-SAMP-076	_classify_decoded in dr_support/services/admission.py	[E4] (automated quality assessment is a separate problem)
IMG-SAMP-080	None	[E7]; [E8]
IMG-SAMP-090	None	[E8]; [E10]
IMG-SAMP-091	docs/reference/DATASET_MANIFEST.md (image_id, image_sha256)	[E10]
IMG-SAMP-092	None	Owner decision (V4)
IMG-SAMP-100	None	Owner decision (V4 acceptance)
IMG-SAMP-101	docs/clinician/IMAGE_SELECTION_GUIDE.html	Nielsen Norman Group, <i>Progressive Disclosure</i>
IMG-SAMP-110	docs/operations/SECURITY_PRIVACY.md; normalize_patient_key in dr_support/services/resolver.py	None

10.4 External sources

The sources below were opened and checked in September 2026. Two sources could not be verified in this pass and are not cited as evidence: the full text of the ICO *Guidelines for Diabetic Eye Care* (2017) and the current AAO *Diabetic Retinopathy Preferred Practice Pattern*.

- [E1] NHS England. *Diabetic eye screening: guidance on fundus image quality and when adequate images cannot be taken*. GOV.UK, updated 11 March 2026. <https://www.gov.uk/government/publications/diabetic-eye-screening-pathway-for-images-and-where-images-cannot-be-taken/diabetic-eye-screening-guidance-when-adequate-images-cannot-be-taken>
- [E2] NHS England. *NHS Diabetic Eye Screening Programme: grading definitions for referable disease*. GOV.UK, updated 4 December 2025. <https://www.gov.uk/government/publications/diabetic-eye-screening-retinal-image-grading-criteria>
- [E3] U.S. FDA. *De Novo decision summary DEN180001 (IDx-DR)*. 2018. https://www.accessdata.fda.gov/cdrh_docs/reviews/DEN180001.pdf
- [E4] Fu H. et al. *Evaluation of Retinal Image Quality Assessment Networks in Different Color-Spaces (EyeQ)*. MICCAI 2019. <https://arxiv.org/abs/1907.05345>
- [E5] NEMA. *DICOM PS3.3, C.8.17.5 Ocular Region Imaged Module and A.41 Ophthalmic Photography 8 Bit Image IOD*. https://dicom.nema.org/medical/dicom/current/output/chtml/part03/sect_C.8.17.5.html
- [E6] U.S. FDA, Health Canada, and MHRA. *Good Machine Learning Practice for Medical Device Development: Guiding Principles*. 2021. <https://www.fda.gov/medical-devices/software-medical-device-samd/good-machine-learning-practice-medical-device-development-guiding-principles>
- [E7] U.S. FDA. *Artificial Intelligence-Enabled Device Software Functions: Lifecycle Management and Marketing Submission Recommendations*. Draft guidance, January 2025. <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/artificial-intelligence-enabled-device-software-functions-lifecycle-management-and-marketing>
- [E8] Alderman J.E. et al. *Tackling algorithmic bias and promoting transparency in health datasets: the STANDING Together consensus recommendations*. *Lancet Digital Health* 2025;7(1):e64–e88. <https://www.datadiversity.org/recommendations>
- [E9] Collins G.S. et al. *TRIPOD+AI statement*. *BMJ* 2024;385:e078378. <https://www.bmj.com/content/385/bmj-2023-078378>
- [E10] Gebru T. et al. *Datasheets for Datasets*. *Communications of the ACM*, 2021. <https://arxiv.org/abs/1803.09010>
- [E11] Tampu I.E., Eklund A., Haj-Hosseini N. *Inflation of test accuracy due to data leakage in deep learning-based classification of OCT images*. *Scientific Data* 2022. <https://www.nature.com/articles/s41597-022-01618-6>